INCH-POUND
MIL-PRF-8805/99J
w/Amendment 1
12 December 2019
SUPERSEDING
MIL-PRF-8805/99J
21 October 2019

PERFORMANCE SPECIFICATION SHEET

# SWITCHES, PUSH, 2 CIRCUIT AND DPDT, ALTERNATE AND MOMENTARY ACTION, 2 AMPERES, SPLASHPROOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-8805.

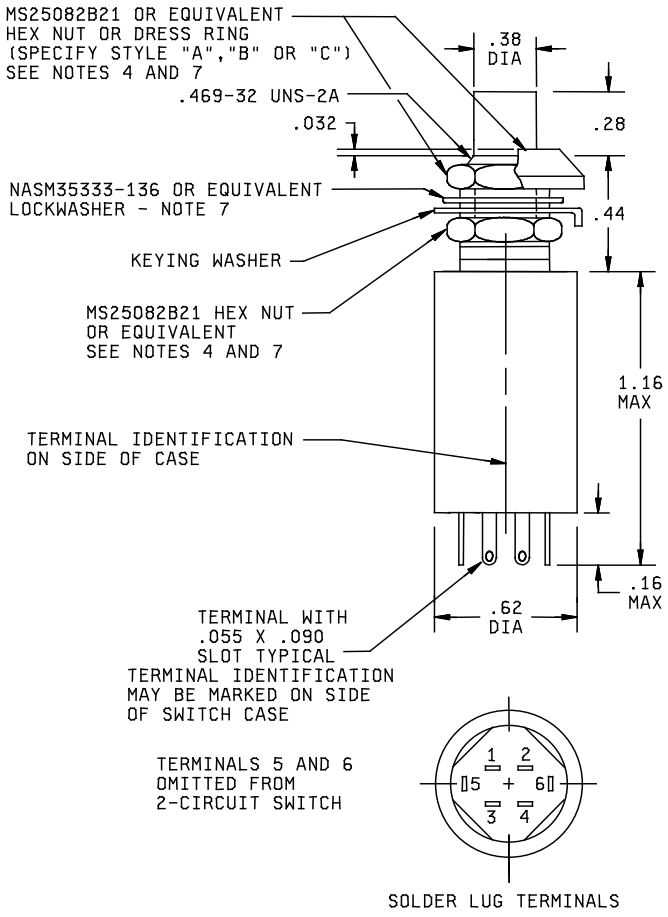
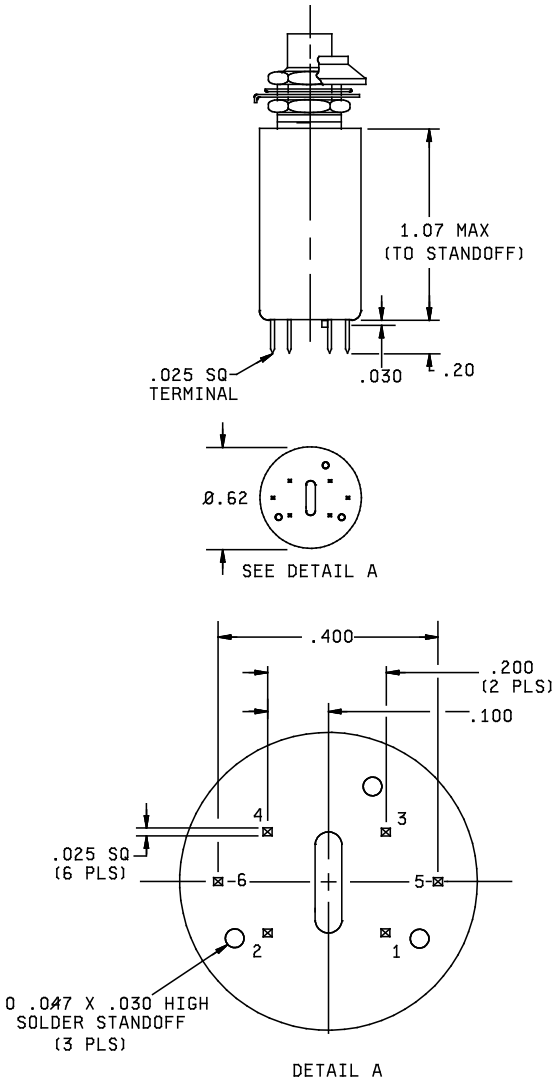



FIGURE 1. Dimensions and configurations.



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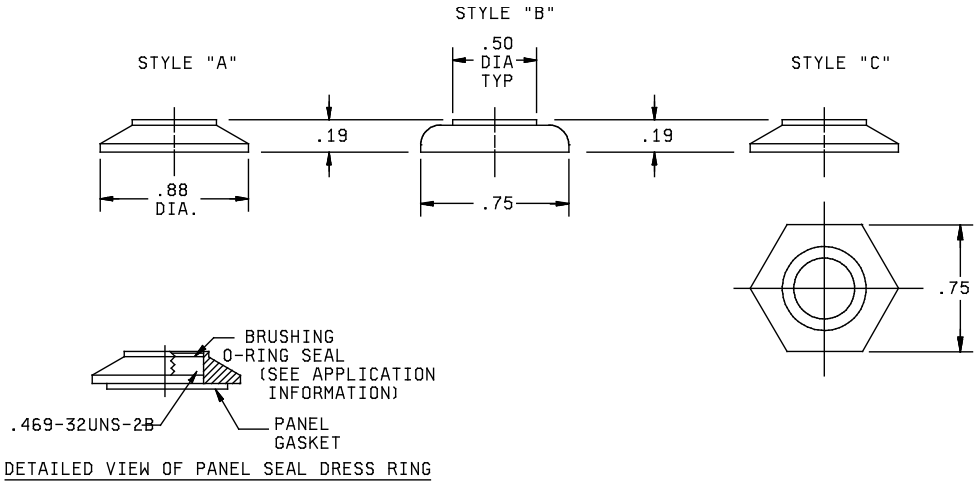

PRINTED CIRCUIT BOARD TERMINAL

| Inches | mm |
|---|---|
| .002 | 0.05 |
| .003 | 0.08 |
| .004 | 0.10 |
| .005 | 0.13 |
| .006 | 0.15 |
| .015 | 0.38 |
| .016 | 0.41 |
| .02 | 0.51 |
| .025 | 0.63 |
| .030 | 0.76 |
| .032 | 0.81 |
| .047 | 1.19 |
| .048 | 1.22 |
| .050 | 1.27 |
| .055 | 1.39 |
| .090 | 2.29 |
| .10 | 2.54 |
| .110 | 2.79 |
| .136 | 3.45 |
| .16 | 4.06 |
| .19 | 4.83 |
| .20 | 5.08 |
| .28 | 7.11 |
| .349 | 8.86 |
| .375 | 9.52 |
| .38 | 9.60 |
| .40 | 10.16 |
| .44 | 11.18 |
| .442 | 11.23 |
| .446 | 11.33 |
| .468 | 11.89 |
| .469 | 11.91 |
| .470 | 11.94 |
| .50 | 12.70 |
| .62 | 15.75 |
| .719 | 18.26 |
| .75 | 19.05 |
| .88 | 22.35 |
| 1.07 | 27.18 |
| 1.16 | 29.47 |

FIGURE 1. Dimensions and configurations - Continued.



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PANEL SEAL DRESS RING

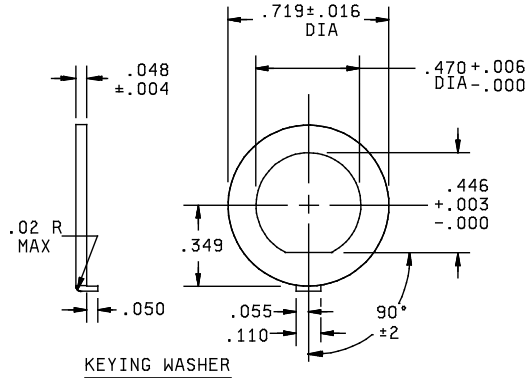
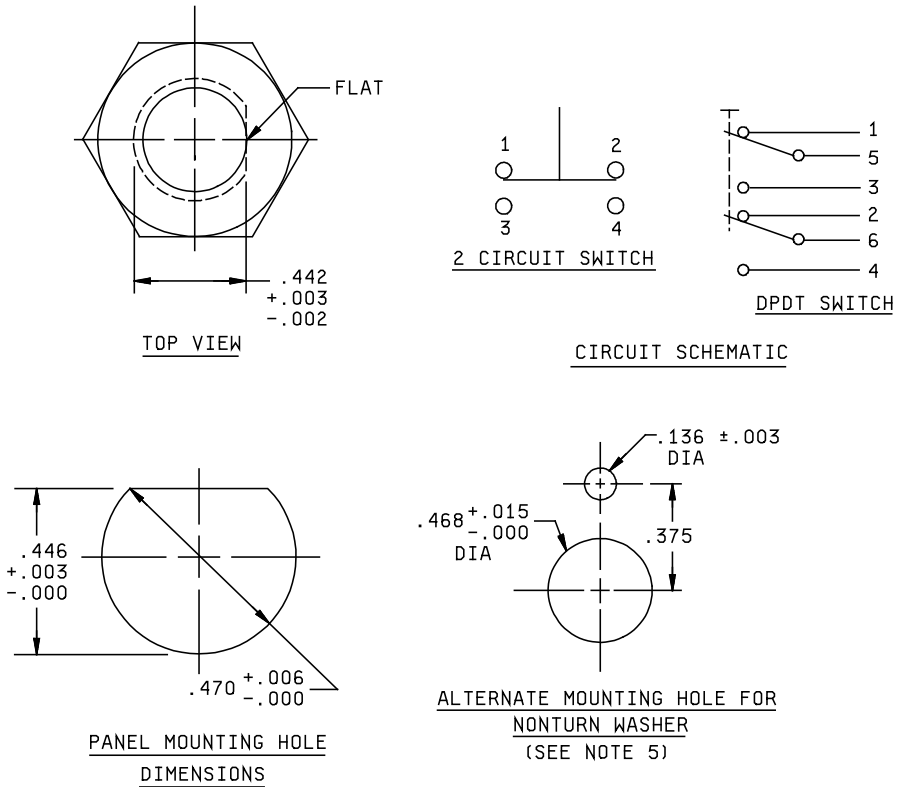



FIGURE 1. Dimensions and configurations - Continued.



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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (0.8 mm) for two place decimals and ±.005 (0.13 mm) for three place decimals.
4. Panel dress ring replaces one hex nut.
5. Alternate mounting hole (2 hole) should not be used for panel seal applications.
6. Terminals 5 and 6 omitted from 2-circuit switch.
7. Alternative base metals and protective finishes as approved by the qualifying activity, may be utilized for hardware material. Dimensions shall be in accordance with the referenced hardware specification.
8. MS25082B21 per MS25082. MS35333-136 per NSAM35333.

FIGURE 1. Dimensions and configurations - Continued.



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REQUIREMENTS:

Dimensions and configurations: See figure 1.

Enclosure design: 6 (splashproof) when panel seal dress ring is specified.

Temperature characteristic: 1 (-55°C to +85°C).

Shock type:

M, except 50 g's, test condition A MIL-STD-202-213: Applicable to all switches.

Type H (high impact), in accordance with MIL-DTL-901, Grade A, Class II. except maximum allowable opening of closed contacts or closing of open contacts is 39 milliseconds. Applicable to momentary action switches only.

Sinusoidal vibration grade: 2 (10 to 2,000 Hz).

Weight: .643 ounce maximum.

Operating characteristics:

Actuating force:..... 3 pounds to 6 ponds.

Plunger travel:.....176 inch maximum.

Dielectric withstanding voltage:

Sea level: 1,000 V rms.

Altitude: 500 V rms.

Terminal strength:

Printed circuit board compatible pin terminals: Applied force shall be 3 pounds minimum.

Mechanical endurance: 50,000 cycles.

Low temperature: Cycling rate of 30 to 65 cycles per minute.

Electrical endurance: 25,000 cycles.

Electrical ratings: See table I.

Qualification: inspection (group submission): See table II.



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Part or Identifying number (PIN): See tables III and IV.

Application information: Bushing O-ring seal (applicable to units with panel seal) has the potential of stretching and rolling-over onto the flat "D" section of the threaded mounting bushing as the dress bezel ring is installed, causing an improper seating of the O-ring. It is recommended that silicone grease be applied to the O-ring seal in the dress bezel prior to installation to prevent this problem.

TABLE I. Electrical ratings.

| Load | Sea level 28 V dc 115 V ac, 400 Hz | 70,000 feet 28 V dc |
|---|---|---|
| | (amperes) | (amperes) |
| Resistive | 2.0 | 2.0 |
| Inductive | 1.5 | 1.5 |
| Lamp | 0.5 | 0.5 |

TABLE II. Qualification inspection (group submission).

| Inspection | Basic Switch | Other switch Samples | Extent of approval |
|---|---|---|---|
| Qualification Inspection Table of MIL-PRF-8805 groups 1 through VIII inclusive | M8805/99-029<br><br>(All sample units) | Two sample units each of M8805/99-006, and -015:<br>Visual and mechanical<br>Vibration<br>Shock<br>Operating characteristics<br><br>Two sample units each of M8805/99-061:<br>Visual and mechanical<br>Solderability<br>Terminal strength<br>Operating characteristics | All |



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TABLE III. PIN and characteristics

| PIN M8805/99- | Dress ring | Switch action | Switch circuit | Button color | Terminals |
|---|---|---|---|---|---|
| 001 | A | Alternate action | 2 circuit | Black | Solder lug |
| 002 | B | Alternate action | 2 circuit | Black | Solder lug |
| 003 | C | Alternate action | 2 circuit | Black | Solder lug |
| 005 | A | Alternate action | 2 circuit | Red | Solder lug |
| 006 | B | Alternate action | 2 circuit | Red | Solder lug |
| 007 | C | Alternate action | 2 circuit | Red | Solder lug |
| 033 | A | Alternate action | 2 circuit | White | Solder lug |
| 034 | B | Alternate action | 2 circuit | White | Solder lug |
| 035 | C | Alternate action | 2 circuit | White | Solder lug |
| 009 | A | Momentary | 2 circuit | Black | Solder lug |
| 010 | B | Momentary | 2 circuit | Black | Solder lug |
| 011 | C | Momentary | 2 circuit | Black | Solder lug |
| 013 | A | Momentary | 2 circuit | Red | Solder lug |
| 014 | B | Momentary | 2 circuit | Red | Solder lug |
| 015 | C | Momentary | 2 circuit | Red | Solder lug |
| 016 | NONE | Momentary | 2 circuit | Red | Solder lug |
| 037 | A | Momentary | 2 circuit | White | Solder lug |
| 038 | B | Momentary | 2 circuit | White | Solder lug |
| 039 | C | Momentary | 2 circuit | White | Solder lug |
| 017 | A | Alternate action | DPDT | Black | Solder lug |
| 018 | B | Alternate action | DPDT | Black | Solder lug |
| 019 | C | Alternate action | DPDT | Black | Solder lug |
| 021 | A | Alternate action | DPDT | Red | Solder lug |
| 022 | B | Alternate action | DPDT | Red | Solder lug |
| 023 | C | Alternate action | DPDT | Red | Solder lug |
| 041 | A | Alternate action | DPDT | White | Solder lug |
| 042 | B | Alternate action | DPDT | White | Solder lug |
| 043 | C | Alternate action | DPDT | White | Solder lug |
| 025 | A | Momentary | DPDT | Black | Solder lug |
| 026 | B | Momentary | DPDT | Black | Solder lug |
| 027 | C | Momentary | DPDT | Black | Solder lug |
| 029 | A | Momentary | DPDT | Red | Solder lug |
| 030 | B | Momentary | DPDT | Red | Solder lug |
| 031 | C | Momentary | DPDT | Red | Solder lug |
| 045 | A | Momentary | DPDT | White | Solder lug |
| 046 | B | Momentary | DPDT | White | Solder lug |
| 047 | C | Momentary | DPDT | White | Solder lug |



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TABLE III. PIN and characteristics - Continued.

| PIN M8805/99- | Dress ring | Switch action | Switch circuit | Button color | Terminals |
|---|---|---|---|---|---|
| 049 | A | Alternate action | 2 circuit | Black | Printed circuit |
| 050 | B | Alternate action | 2 circuit | Black | Printed circuit |
| 051 | C | Alternate action | 2 circuit | Black | Printed circuit |
| 053 | A | Alternate action | 2 circuit | Red | Printed circuit |
| 054 | B | Alternate action | 2 circuit | Red | Printed circuit |
| 055 | C | Alternate action | 2 circuit | Red | Printed circuit |
| 057 | A | Alternate action | 2 circuit | White | Printed circuit |
| 058 | B | Alternate action | 2 circuit | White | Printed circuit |
| 059 | C | Alternate action | 2 circuit | White | Printed circuit |
| 061 | A | Momentary | 2 circuit | Black | Printed circuit |
| 062 | B | Momentary | 2 circuit | Black | Printed circuit |
| 063 | C | Momentary | 2 circuit | Black | Printed circuit |
| 065 | A | Momentary | 2 circuit | Red | Printed circuit |
| 066 | B | Momentary | 2 circuit | Red | Printed circuit |
| 067 | C | Momentary | 2 circuit | Red | Printed circuit |
| 069 | A | Momentary | 2 circuit | White | Printed circuit |
| 070 | B | Momentary | 2 circuit | White | Printed circuit |
| 071 | C | Momentary | 2 circuit | White | Printed circuit |
| 073 | A | Alternate action | DPDT | Black | Printed circuit |
| 074 | B | Alternate action | DPDT | Black | Printed circuit |
| 075 | C | Alternate action | DPDT | Black | Printed circuit |
| 077 | A | Alternate action | DPDT | Red | Printed circuit |
| 078 | B | Alternate action | DPDT | Red | Printed circuit |
| 079 | C | Alternate action | DPDT | Red | Printed circuit |
| 081 | A | Alternate action | DPDT | White | Printed circuit |
| 082 | B | Alternate action | DPDT | White | Printed circuit |
| 083 | C | Alternate action | DPDT | White | Printed circuit |
| 085 | A | Momentary | DPDT | Black | Printed circuit |
| 086 | B | Momentary | DPDT | Black | Printed circuit |
| 087 | C | Momentary | DPDT | Black | Printed circuit |
| 089 | A | Momentary | DPDT | Red | Printed circuit |
| 090 | B | Momentary | DPDT | Red | Printed circuit |
| 091 | C | Momentary | DPDT | Red | Printed circuit |
| 093 | A | Momentary | DPDT | White | Printed circuit |
| 094 | B | Momentary | DPDT | White | Printed circuit |
| 095 | C | Momentary | DPDT | White | Printed circuit |



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Table IV. Cancelled PIN's.

| M8805/99- | M8805/99- | M8805/99- | M8805/99- |
|---|---|---|---|
| 004 | 028 | 052 | 076 |
| 008 | 032 | 056 | 080 |
| 012 | 036 | 060 | 084 |
| 020 | 040 | 064 | 088 |
| 024 | 044 | 068 | 092 |
| | 048 | 072 | 096 |

Reference Documents. In addition to MIL-PRF-8805, this document references the following:
MIL-DTL-901
MIL-STD-202-213
MS25082
NASM35333

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

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